AIN-3009 MLOps Term Project

Predictive Maintenance

End-to-End ML Lifecycle on Azure ML + MLflow

A complete MLOps lifecycle project: from dataset and experiments to registry, deployment, and monitoring.

Course
AIN-3009 Delivering AI Applications with MLOps

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Dataset — **Training** — **Tuning** — **Registry** — **Endpoint** — **Monitoring**

3.4%	**0.974**	**v1**	**Succeeded**
failure rate	best CV ROC-AUC	registered model	endpoint state

5-minute technical project presentation

Project Objective

What the assignment asks us to demonstrate

1 **Experiment tracking**
Log parameters, metrics, artifacts, and outputs with MLflow.

2 **Model training and tuning**
Train baselines and tune hyperparameters with traceable runs.

3 **Model deployment**
Package the model and serve predictions through an endpoint.

4 **Performance monitoring**
Track model behavior over time with drifted batches.

5 **Model registry**
Version the model and manage Staging → Production lifecycle.

Project framing

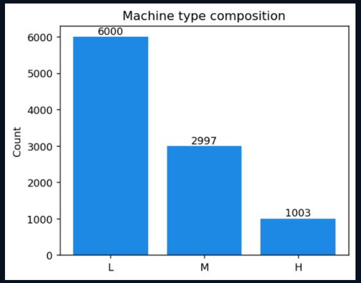
The rest of the deck maps each requirement to a concrete project artifact: code, metrics, MLflow logs, Azure registry, endpoint evidence, and monitoring charts.

Dataset & Problem Framing

AI4I 2020 predictive-maintenance data: imbalanced binary classification

What data did we use?

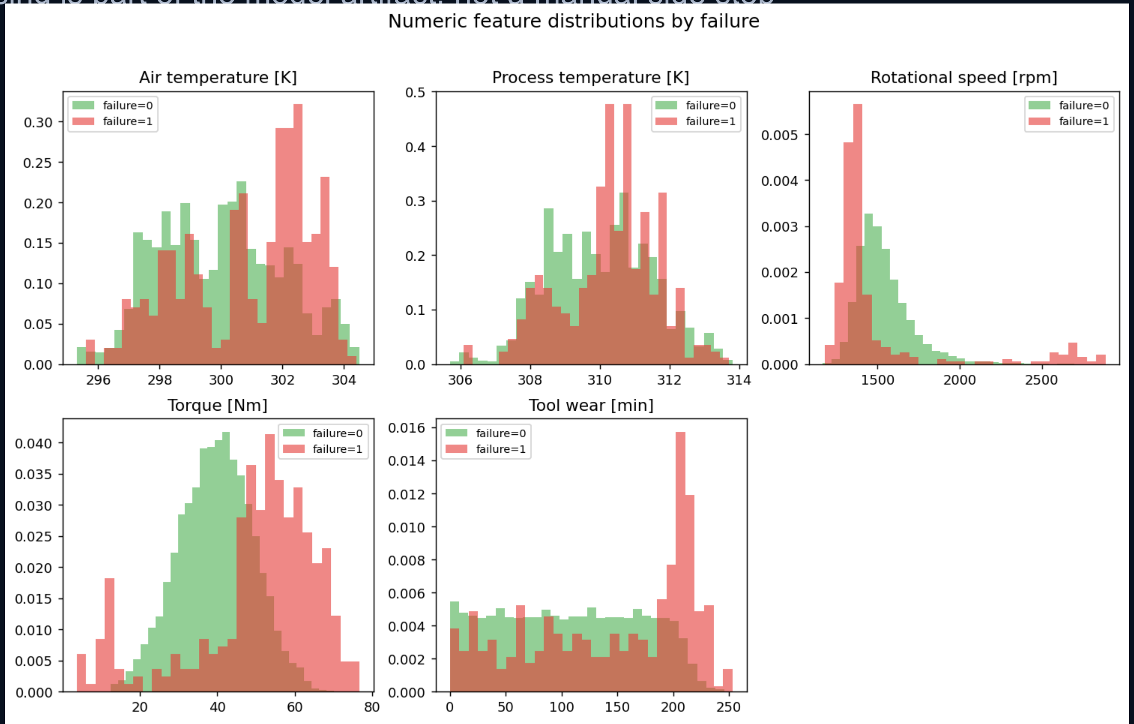
- Source: AI4I 2020 Predictive Maintenance dataset
- 10,000 machine telemetry records
- Target: Machine failure (0/1)
- Failure rate is only 3.4%, so accuracy is misleading
- Leakage columns removed: TWF, HDF, PWF, OSF, RNF

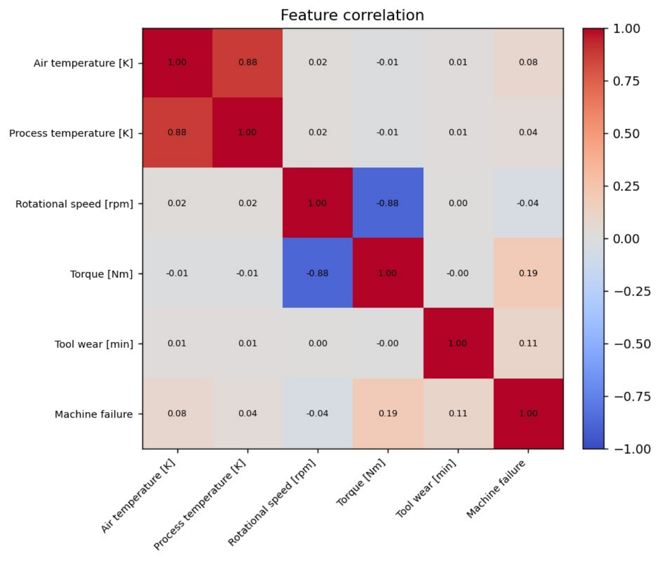


Evaluation focus: ROC-AUC, PR-AUC, recall, and F1 instead of raw accuracy.

Data Preparation & EDA

Preprocessing is part of the model artifact, not a manual side step





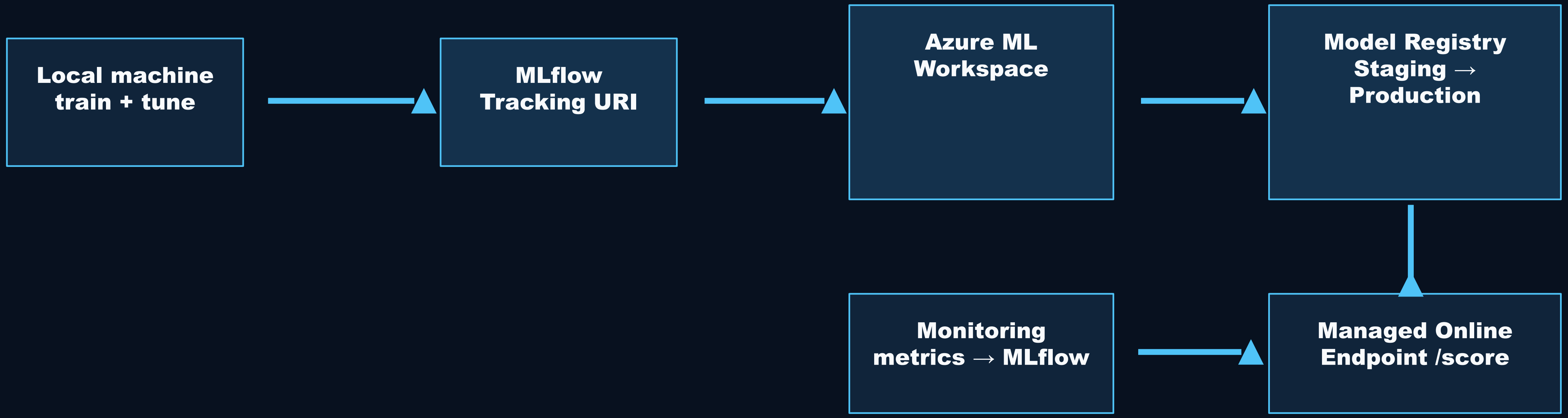
The scikit-learn Pipeline serializes preprocessing with the estimator, so training and serving use the same transformations.

Preprocessing pipeline

- Numeric features → StandardScaler
- Machine Type → OneHotEncoder

Architecture: Local Training, Cloud Lifecycle

Azure credit is used for shared MLOps infrastructure, not expensive training compute

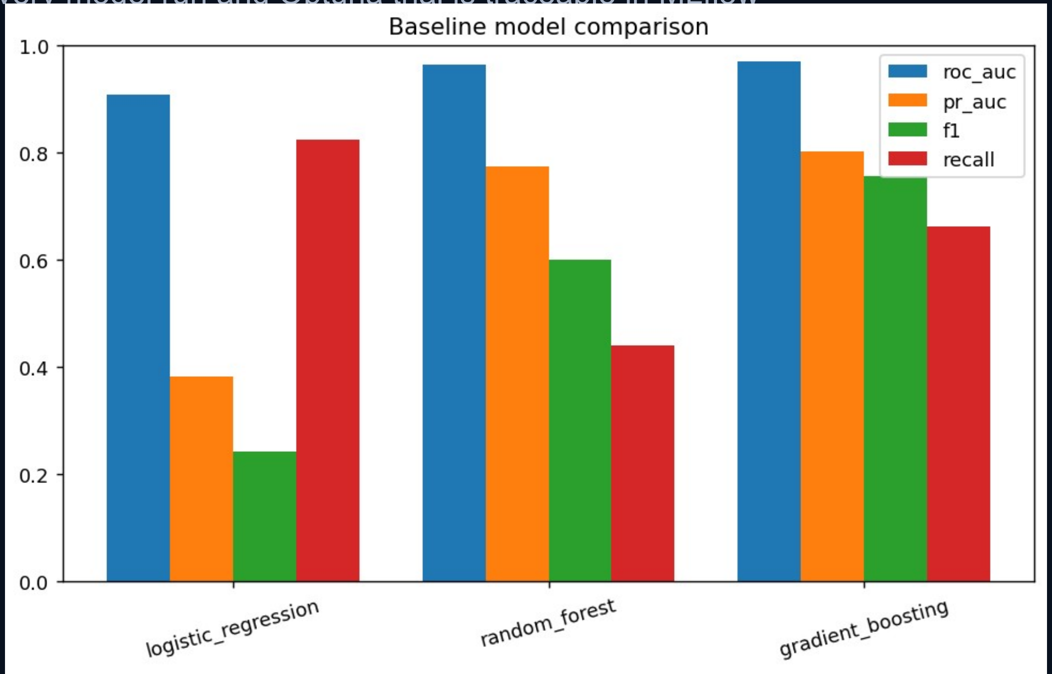


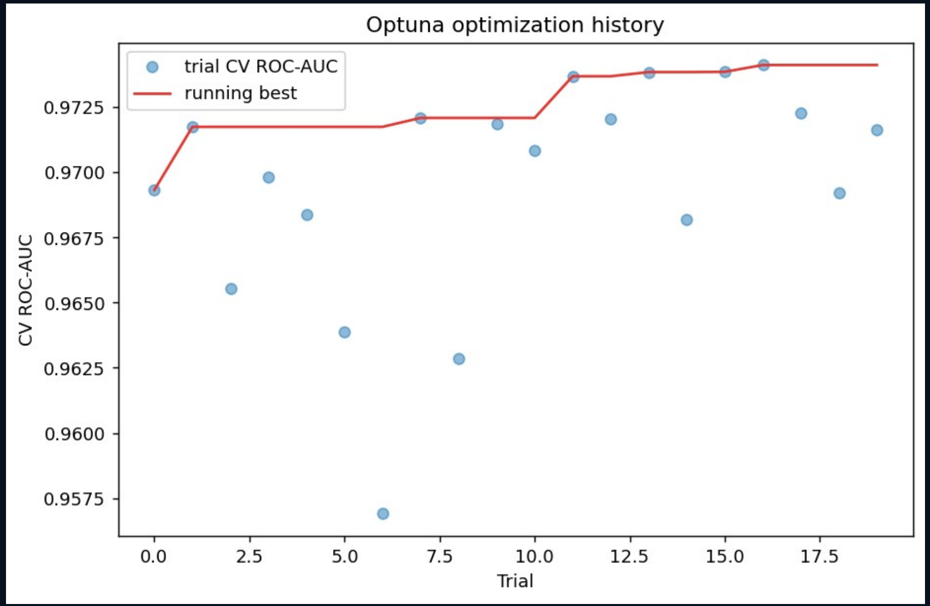
Stores: experiments, parameters, metrics, artifacts, registered models, and endpoint state.

Tracking | Training/Tuning | Registry | Deployment | Monitoring

Training, Tracking & Tuning

Every model run and Optuna trial is traceable in MLflow





20
Optuna trials

0.974
Best CV ROC-AUC

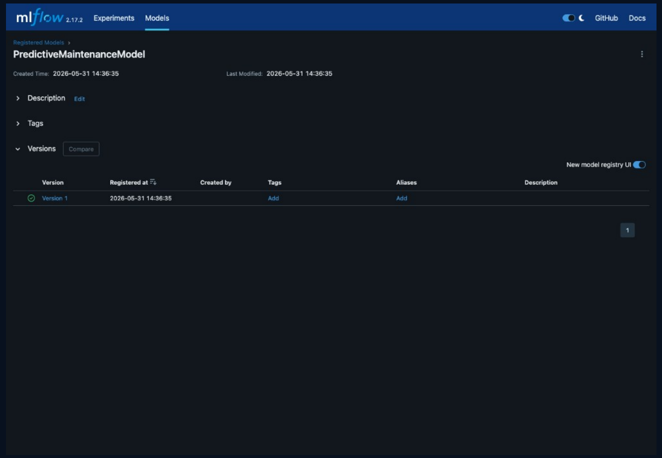
0.44 → 0.77
Recall lift

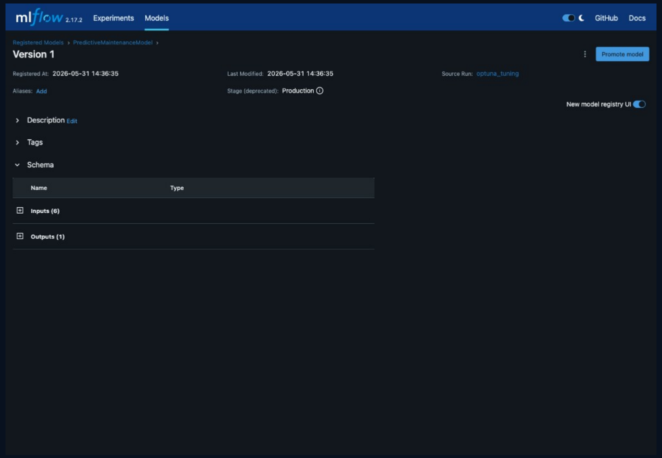
MLflow logs

- parameters, metrics, classification report, confusion matrix, serialized model artifact

Registry, Deployment & Pipeline Automation

The model becomes a versioned, deployable asset





PredictiveMaintenanceModel

Registered model

1

Version

Staging → Production

Lifecycle

Succeeded

Endpoint state

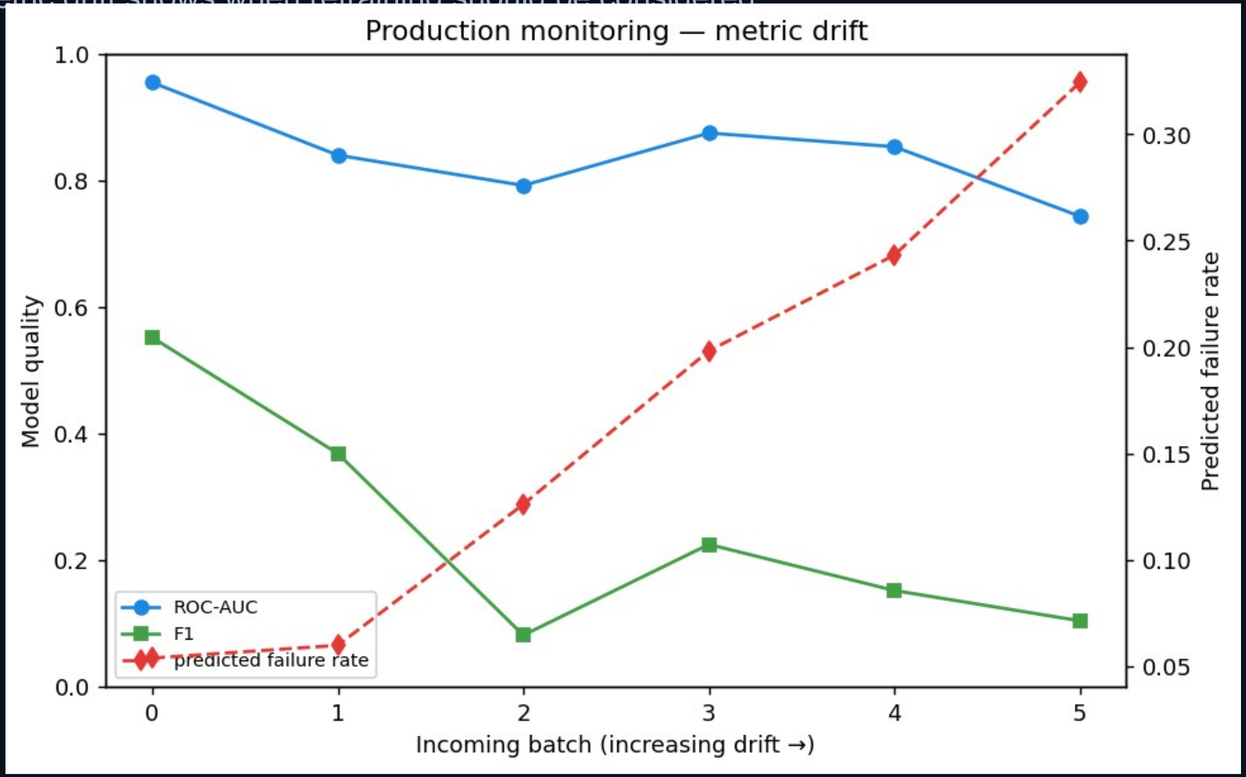
Airflow DAG orchestration

validate — train — tune — register — monitor

Tracking | Training/Tuning | **Registry** | Deployment | Monitoring

Monitoring Closes the MLOps Loop

Metric drift shows when retraining should be considered



0.914
Batch 0 ROC-AUC

0.766
Batch 5 ROC-AUC

4% → 37%
Predicted failure rate

Course objectives covered

- ✓ **Experiment tracking**
- ✓ **Training and tuning**
- ✓ **Registry and deployment**
- ✓ **Monitoring and drift tracking**

Final takeaway: this is not a notebook-only model; it is an end-to-end MLflow + Azure ML lifecycle demonstration.